

The Wiley Blackwell Companion to Phonology, Second Edition

Entry title: The Representation of Intonation

First author: Full name and affiliation; plus email address if corresponding author
Amalia Arvaniti, Radboud University; amalia.arvaniti@ru.nl

To appear in: Kuniya Nasukawa, Bridget Samuels, Geoff Schwartz, Miklós Törkenczy (eds.), *The Wiley Blackwell Companion to Phonology* (2nd ed). Wiley Blackwell.

Word Count

9913 words

Abstract

Two approaches dominate the study of intonation with respect to its representation: tunes are treated either as configurations or as composites of primitives (which are either configurations themselves, such as rises and falls, or level tones, such as highs and lows, and composites thereof). Here, tracing the historical provenance of these two approaches shows that many relevant observations go as far back as the early 20th century. Additionally, the essential tenets of each approach are presented together with relevant empirical evidence. This evidence overwhelmingly confirms that configurational approaches are not observationally adequate. On the other hand, treating tunes as composites of primitives which independently contribute both to the tune's form and to intonation meaning offers significant advantages: it is observationally adequate, facilitates cross-linguistic research, and allows researchers to capture important generalizations that help predict the form and understand the pragmatic function of tunes. The compositional approach is best exemplified by the autosegmental-metrical theory of intonational phonology, the main principles of which are presented in some detail, together with an overview of recent developments that have led to a better understanding of the relationship between phonological representation and intonation phonetics.

Keywords

intonation, F0, pitch, prosody, autosegmental-metrical, compositionality, configuration

Main text

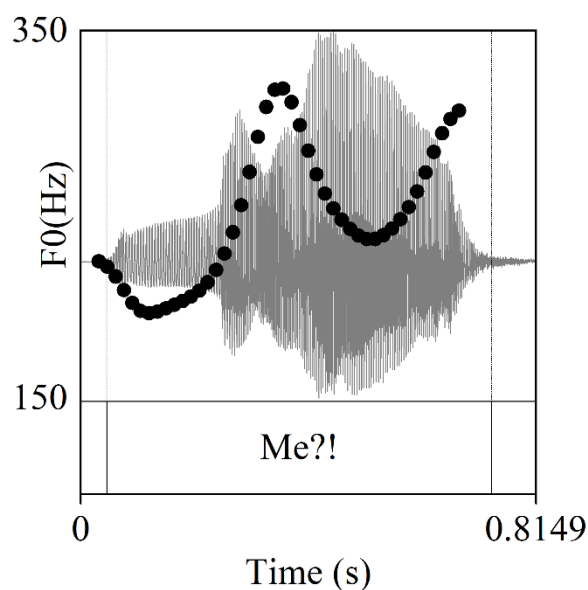
1. Introduction

All utterances, in all languages, show pitch modulations. These modulations are due to changes in fundamental frequency (F0), the physical property of the speech signal that is perceived as pitch and is largely determined by the rate of vocal fold vibration. Pitch modulations during speaking are determined by both linguistic and paralinguistic factors (Ladd 2008, ch. 1). This paper covers the former, but does not address paralinguistic uses of pitch to indicate excitement, anger, fatigue and the like.

The phonological provenance and function of linguistic uses of F0 varies crosslinguistically. In many languages, pitch specifications are determined at the lexical level and change the meaning of words. This applies to *tone languages* (such as Mandarin, Thai, and Igbo), in which most syllables are lexically specified for tone, and to *languages with lexical pitch accent* (such as Japanese, Swedish, Limburgish, Serbian, Western Basque, and Yaqui), in which tone operates in a similar fashion, except that at most one syllable per word is lexically specified for tone and many words may not have a tonal specification at all. Additionally, all languages, including those that have tone or lexical pitch accent, show pitch modulations that are specified at the phrase (postlexical) level. The term *intonation* refers to these phrase level pitch modulations which are particularly noticeable in languages without a lexical tonal component. In such languages (e.g. English, French, Greek, Korean), the entire F0 contour is specified at the phrase level by means of a complex interplay between metrical structure, prosodic phrasing, syntax, and pragmatics. These factors determine where in the utterance pitch movements will occur and of what type they will be; e.g. when uttered as a piece of news, *Melinda saw a platypus* will have pitch peaks on *Melinda* and *platypus*, but when the same phrase is used to correct the subject's identity (*MELINDA saw a platypus, not Jack*), only *Melinda* will have a pitch peak.

Determining the structure and representation of intonation can be challenging, since the realization of tunes is dependent on the segmental material with which tunes co-occur, while their meaning is pragmatic and interpreted probabilistically in context (Hirschberg, 2004; see §4.3). The following examples illustrate these points. In Figure 32.1, *Me?! and A ballgown designer?! are both plausible responses to a career advisor's pronouncement that, according to test results, designing ballgowns is the recommended career choice for a student who had always dreamt of becoming an aerospace engineer (for similar examples, see Ladd 2008: 45–46; Arvaniti 2022). The short contour can be described as a rise–fall–rise, but the longer contour includes a low F0 stretch between the rise–fall and the final rise, and thus the same description does not apply. Despite their differences in form, in American English, both contours convey incredulity (Ward and Hirschberg 1985; Hirschberg and Ward 1992), and thus should be treated as instances of the same tune.*

(a)



(b)

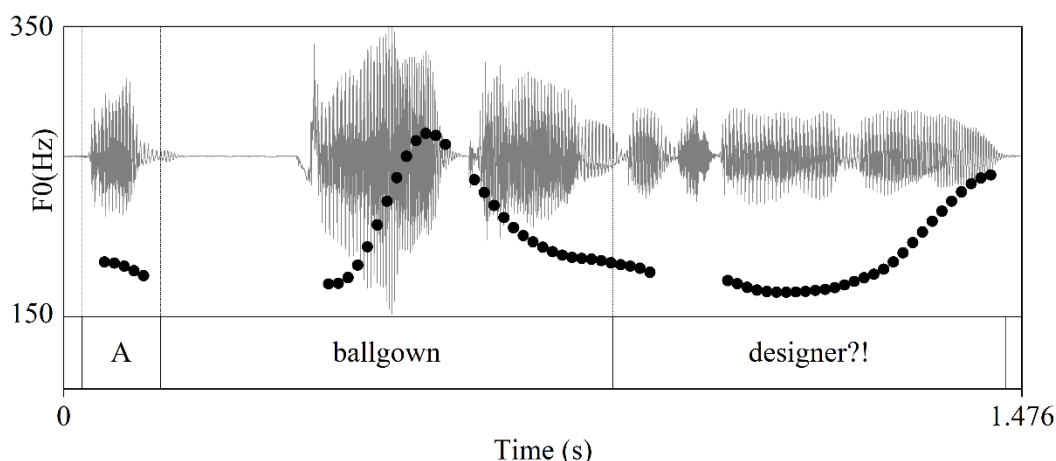
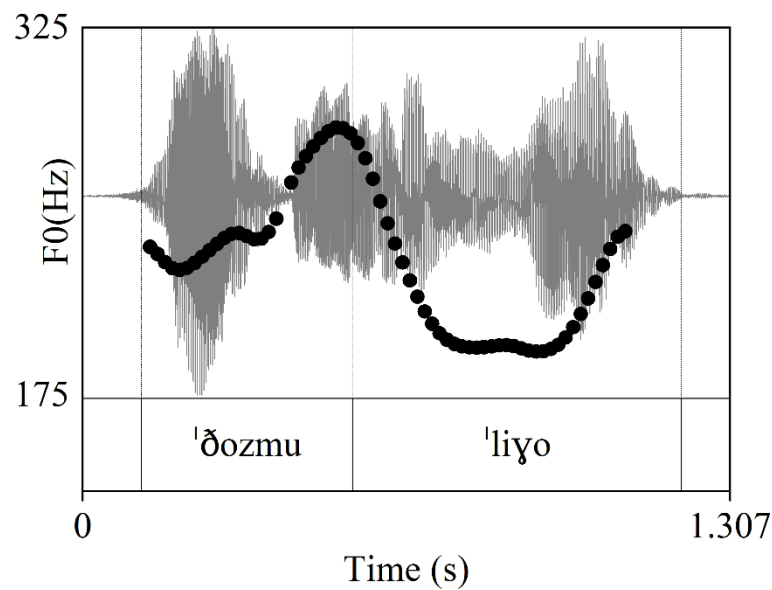


Figure 32.1 Waveforms and F0 contours of two English utterances illustrating the incredulity contour (Hirschberg and Ward 1992); in (a), *Me?!*; in (b), *A ballgown designer?!*. Vertical lines indicate word boundaries. The images are sized to reflect differences in utterance length.

Figure 32.2, shows Greek pitch contours very similar to those in Figure 32.1 and presenting comparable changes with respect to utterance length. This comparability indicates that the same principles apply cross-linguistically to intonation form. With respect to pragmatic function, on the other hand, we observe the following. First, while the English tune expresses incredulity, the Greek tune is used with negative statements containing the negative operator [ðen] “not”, polite imperatives, and wh-questions (Arvaniti and Baltazani 2005; Baltazani 2006a; Arvaniti and Ladd 2009; Baltazani, Gryllia, and Arvaniti 2020). These function differences between highly comparable tunes makes it obvious that there is no “natural” universal meaning for a given tune: no Greek speaker would use the tune discussed here to indicate incredulity, and no English speaker would use it to ask a question. Further, the fact that the Greek tune is used in two distinct pragmatic contexts

supports the view that a tune can serve various pragmatic purposes within a linguistic system (cf. Arvaniti, Gryllia and Baltazani, to appear).

(a)



(b)

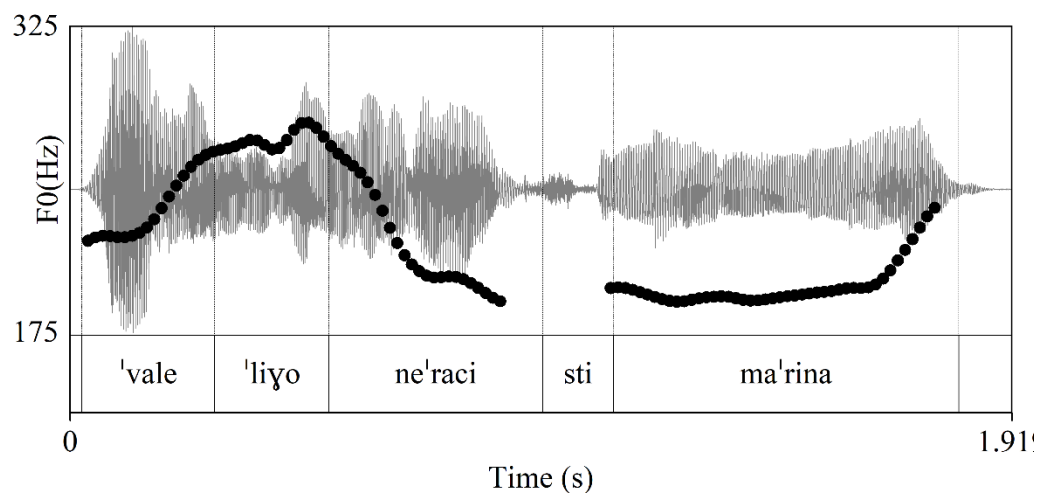
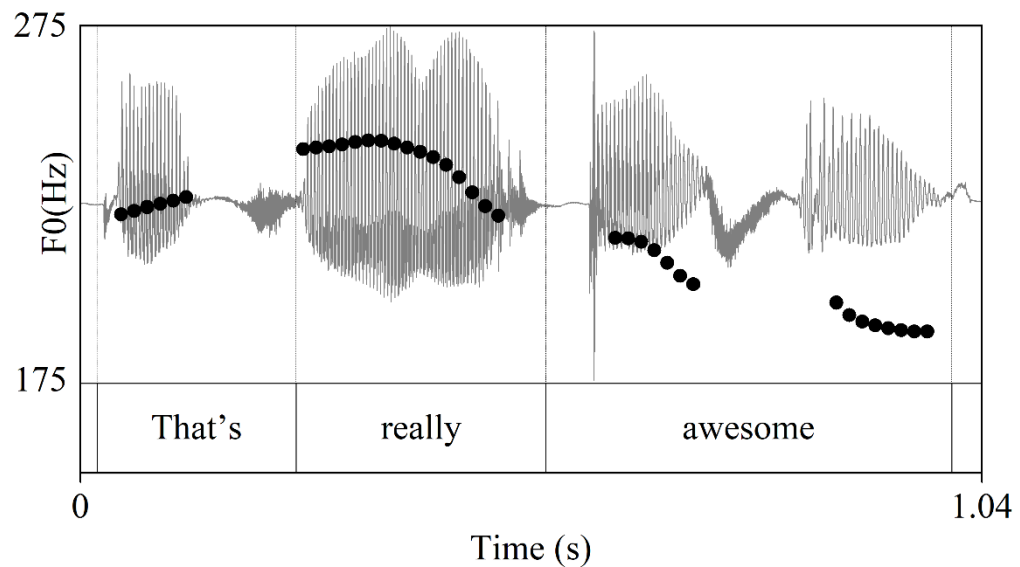


Figure 32.2 Waveforms, transcriptions, and F0 contours of two Greek imperatives used for polite requests; in (a), ['ðozmu 'liyo] 'give me some'; in (b), ['vale 'liyo ne'raci sti ma'rina] 'give some water to Marina'. Vertical lines indicate word boundaries. The images are sized to reflect differences in utterance length.

Figure 32.3 illustrates two instances of another English tune: unlike the contours in Figures 32.1 and 32.2, which look different from each other but convey the same meaning in each case, the contours in Figure 32.3 are realizations of the same American English tune but interpreted differently. In both *That's twenty dollars* and *That's really awesome*, the two content words show pitch peaks the second of which is *downstepped* (significantly lower than the first). In *That's twenty dollars*, this

relationship between peaks is the default, as *dollars* in a US context is the expected currency. When the tune is used with *That's really awesome*, however, the downstepped pitch peak on *awesome* makes it sound routine and predictable, suggesting that the speaker is being sarcastic or indifferent.

(a)



(b)

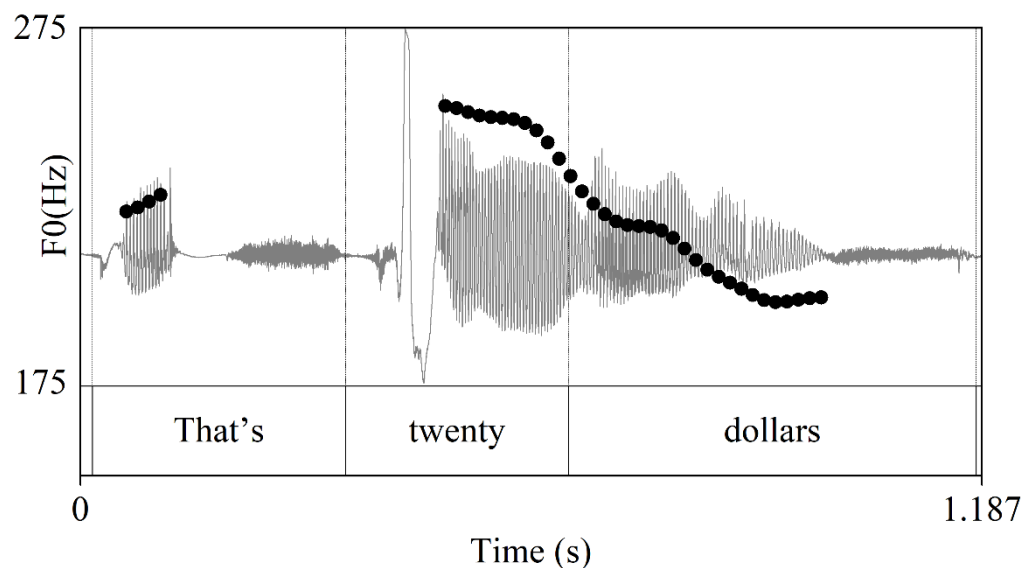


Figure 32.3 Waveforms and F0 contours of two English utterances; in (a), *That's really awesome*; in (b), *That's twenty dollars*. Vertical lines indicate word boundaries.

These examples illustrate three main points about intonation. First, the shape of a tune with a given pragmatic interpretation can vary depending on the segmental material with which the tune is co-

produced. Such differences are not random, but related to the length and metrical structure of the utterance with which the tune is associated, including the number of syllables, phrasing, and, for languages with stress, the location of stresses (for extensive discussions of context dependent variation see Arvaniti and Ladd 2009, Arvaniti 2016, Arvaniti, Katsika, and Hu 2024). Second, tunes as a whole do not have a constant meaning, either within or across languages: within a language, their interpretation depends on lexical choices and pragmatic context; across languages differences can be arbitrary. A successful theory of intonation should be able to capture these properties, explain the connection between intonation and meaning and make generalizations from surface F0 data with sufficient predictive power to generate new contours of the same basic melody to “fit” new utterances of varying lengths and metrical structures.

In the remainder of the paper, I discuss the solutions proposed to address some of these observations. I first review approaches to intonational primitives, the components of tunes (§2), as debates about them pertain to all models. I then review approaches that treat contours as gestalts or *configurations* (§3), finally covering compositional approaches (§4). In §4.2, I focus on the *autosegmental-metrical theory of intonational phonology* (henceforth *AM*), covering its main tenets and reviewing recent developments.

2. Intonation primitives

Over the years, both dynamic tones, such as rises and falls, and level tones, such as highs and lows, have been used as intonation primitives, i.e. as components of larger tunes, while hybrid models have also been developed.

Early British systems relied largely on dynamic elements (Jones 1909, Armstrong and Ward 1926), namely rises and falls (see §3.1 for details). The models of the American structuralists, on the other hand, were based on level tones, extra-high, high, mid, and low (Pike 1945; Trager and Smith 1951; Hockett 1955). In these systems, level tones were treated as phonological abstractions equivalent to phonemes and were defined relative to each other (rather than as absolute levels). Structuralist analyses were criticized, most notably by Bolinger (1951), whose empirical evidence led him to argue that level tones cannot not be relative but must “rove each in its own bailiwick” (1951: 200). He considered this position untenable in practice, as speaker differences in pitch range would mean that a given pitch could reflect different phonological level tones.

Finally, many systems include both level and dynamic elements. Rises, falls and plateaux are the primitives of the model developed by the Dutch Institute for Perception Research (abbreviated to IPO in Dutch; see ‘t Hart, Collier, and Cohen 1990 and references therein). Similarly, in many models of the British school, some elements are described by means of movements, e.g. *rising nucleus*, and others by means of levels, e.g. *high head* (see §3.1, and Nolan 2022 for an overview). A similarly mixed approach to primitives is taken in INTSINT (*International Transcription System for Intonation*; Hirst and Di Cristo 1998). The same applies to PENTA (*Parallel Encoding and Pitch Target Approximation*), in which each syllable has a specification for F0; these specifications are both level and dynamic elements (see Xu, Prom-on, and Liu 2022 for a review).

Often the choice of primitives is not explicitly motivated and the issue has sometimes been considered mundane: for instance, Bolinger (1986: 225) remarked that “it makes no difference, in describing a movement, whether one says ‘first you are going to be up and then you are going to be down’ or ‘you are going to go down.’” Research, however, suggests that listeners interpret F0 movements as levels (Dilley and Brown 2007). Additionally, conceiving of tunes as sequences of level tones better explains observed regularities in production. Pierrehumbert and Beckman (1988: chs 2–

4) showed that the context-related pitch level differences of Japanese tones are better accounted for using level tones as primitives. Arvaniti, Ladd, and Mennen (1998) tested the predictions of IPO and AM, by examining the realization of Greek rising pitch accents (pitch movements realized on stressed syllables; see §4.2). They found that the timing, duration, and rate of change of these rises were not invariable, as advocated by the IPO. Rather, it was the timing and scaling (F0 level) of the rise onset and offset that were held constant: the initial low F0 coincided with the onset of the accented syllable, while the pitch peak was reached early into the first post-accentual vowel. Arvaniti et al. (1998) interpreted these findings as evidence against dynamic tones as intonation primitives, since the rises they examined did not themselves exhibit invariable features, only their low and high ending points did.

In the past 25 years or so the use of level tones as intonation primitives, at least at the phonological level, has been well established (see §4.2). Nevertheless, it has also become increasingly recognized that, phonetically, groupings of (typically two) tones may enter into different types of relationships: tones may be timed with respect to each other, forming closely knit units more accurately described as dynamic tones; alternatively, the tones may be timed with respect to different segments, leading to a looser connection between them. The pattern reported by Arvaniti et al. (1998) is of the latter type (see also Dilley, Ladd, and Schepman 2005 for similar results on English). Frola (2002), on the other hand, provides evidence from European Portuguese that both types may co-exist in the same language (see Jun and Fletcher 2014 for a discussion). Overall, the evidence indicates that the nature of tune components is not critical in understanding intonation. On the other hand, the conceptualization of the tunes themselves as *gestalts* or compositions of smaller elements is consequential. I turn to these two approaches to tunes next.

3. Configurational approaches

3.1 Configurational approaches and intonation form

Pitch contours have often been treated as *gestalts* or configurations. This applies particularly to older approaches, including Jones (1909, 1932) and Bolinger (1951). It is exemplified by two well-known systems, the British school of intonation (see Nolan 2022 for an overview) and IPO ('t Hart et al. 1990), as well as superpositional models, such as that developed by Fujisaki and colleagues (Fujisaki and Hirose 1984). Configurations, in various forms, are also espoused by some 21st century models, such as OXIGEN (*Oxford Intonation Generator*; Grabe, Kochanski, and Coleman 2003) and the bestiary of English intonation contours of Goodhue et al. (2015).

In the British school, tunes are decomposed into the *pre-head*, *head*, *nucleus*, and *tail*. The nucleus, the pitch movement on the stressed syllable of the most important word of the utterance, is the only required element, and it is treated as a configuration. Any unstressed syllables that precede the first stressed syllable form the pre-head; the F0 from the first stressed syllable to the nucleus forms the head, and any unstressed syllables following the nucleus form the tail. As can be surmised, not all tunes include all four components, while heads in particular can span arbitrary lengths; e.g. in *next week*, the head is only one syllable long (*next*), but in *how many platypi did you see?* it spans eight syllables (*how many platypi did you*).

In the IPO model, the rises and falls that are the model's primitives combine into larger configurations or *contours*; e.g. a rise–plateau–rise, a sequence of three pitch movements, creates a “hat pattern” while a rise–fall creates a “pointed hat” ('t Hart et al. 1990 and references therein).

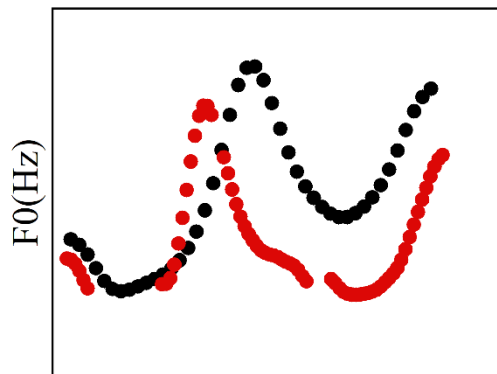
In superpositional models, on the other hand, pitch contours are analysed as consisting of two elements, a general F0 trend and local perturbations which “ride” on it. This conceptualization was

espoused by Bolinger (1986), who used *accentuation* to refer to pitch movements (*accents*) on stressed syllables, and *intonation* to refer to the general course of F0. In this respect, the IPO system can be construed as a superpositional model, in that its primitives and contours ride on top of an automatically declining component ('t Hart et al. 1990: ch. 5). Superpositional models have also been developed for Danish (see Grønnum 2022 for an overview) and Swedish (Gårding 1983). A popular superpositional model is that of Fujisaki. This model was originally developed for Japanese (Fujisaki and Hirose 1984), a language with lexical pitch accent, and has since been used for several other languages, most recently by Torres et al. (2014) who adapted it to English, Spanish, and German. Fujisaki's approach involves a phrase command resulting in a rising–falling F0 course spanning an utterance (or part thereof), and accent commands resulting in localized perturbations. As Ladd (2008: 23 ff.) and Arvaniti and Ladd (2009) point out, however, while this method captures F0 patterns found in Japanese, other Fujisaki-based models resort to counterintuitive solutions – such as negative accent commands and phase commands that span linguistically arbitrary stretches of speech – in order to adequately describe the course of F0 in languages other than Japanese. To address these issues, recent Fujisaki-based models, such as Torres et al. (2014), incorporate in their modelling assumptions that stem from AM, a compositional model.

The popularity of configurational approaches is partly due to the fact that they seem intuitive. As Bolinger (1951: 206) put it, F0 contours can be seen as “a pattern [. . .] in the fundamental, down-to-earth sense of a continuous line that can be traced on a piece of paper”, leading him to argue that “intonation could not be a more appropriate illustration of the Gestalt.” (p. 208). Further, the relationship between shape and function can seem sufficiently natural that it has been argued to derive from the biological underpinnings of pitch production (e.g. Ohala 1983; Gussenhoven 2004: ch. 5; see Calhoun et al. 2024 for evidence that such connections are not biological but socially determined).

Despite their appeal, configurational approaches face major challenges with form: if melodies (or large parts thereof) are undivided wholes, they should simply shrink and stretch to “fit” the duration of the utterance they co-occur with; as argued in Jones (1972: 279), contours should “spread over a large number of syllables, or [. . .] be compressed into smaller spaces.” However, there is overwhelming evidence that tunes are not realized in this accordion-like manner. Rather, as 't Hart et al. (1990, ch. 4) observed in production and confirmed by means of perception experiments (see §3.2), sequences of pitch movements sometimes appear on a single syllable and sometimes are separated by several syllables, without these changes affecting the melodic identity of the contour (for additional evidence see Rietveld and Gussenhoven 1995 on Dutch, D'Imperio and House 1997 on Neapolitan Italian, and Redi 2003 on American English). This critical point is illustrated in Figure 32.4 which shows that the contour of the longer utterance is not a stretched-out copy of the short contour (Figure 32.4a), nor is the short contour a compressed version of the longer one (Figure 32.4b; for further discussion, see Arvaniti and Ladd 2009, and Arvaniti et al. 2024). The inability of configurational approaches to deal with this difference is evident in British school analyses, in which the tune of *Me?!* is analysed as a simple fall-rise, because the entire pitch movement is realized on one syllable, while that of *A ballgown designer?!* is a compound fall-rise, since the fall and the rise appear on different words (Halliday 1970; O'Connor and Arnold 1973). The compound tune is problematic as it exceptionally contains two nuclei (fall and rise) but cannot be distinguished from the simple fall-rise on the basis of meaning.

(a)



(b)

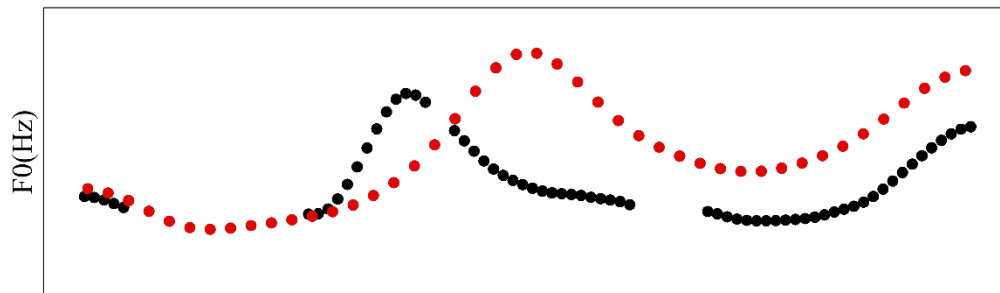


Figure 32.4 The F0 contours of the utterances in Figure 32.1; the red lines represent compressed (a) and stretched-out (b) versions of the long contour in (b) and the short contour in (a), respectively. The images are sized to reflect differences in utterance length.

In short, configurational approaches cannot generalize because they cannot apply the same representation to contours that are superficially different but treated by speakers as instances of the same tune. By the same token, configurational approaches cannot account for, let alone predict, systematic differences between realizations. Thus, they fail one of the main criteria for an adequate theory of intonational phonology mentioned in §1.

3.2 Configurational approaches and intonation meaning

Some configurational models do not explicitly address intonation meaning. For example, in the IPO approach, meaning is said to play no part in establishing either the primitives or the contours themselves because “intonation features have no intrinsic meaning” (’t Hart et al. 1990: 110). The heuristic used is the *close-copy technique*: listeners compare natural and stylized versions of contours until they report that the two do not sound the same. Variants listeners distinguish are used to establish the elements of the intonational system, while similarity helps determine the limits within which contours can phonetically vary.

In configurational models that rely on meaning, such as the British school, meaning is connected to tunes, especially nuclei. However, as illustrated in §1, the relationship between tune and meaning is not one-to-one: the same tune can lend different pragmatic nuances to different utterances, while the same meaning can be expressed by superficially different-looking contours that are realizations of the same underlying tune. Such observed mismatches inevitably lead to disagreements in the analysis of both meaning and form. For instance, Halliday (1970) and Crystal (1972) recognize one type of fall in British English intonation, while O'Connor and Arnold (1973) and Cruttenden (1997) distinguish high and low falls, on the grounds that they reflect different meanings. Additionally, British school meaning descriptions can be vague and contradictory: O'Connor and Arnold (1973: 78–79) describe the “Jackknife” (simple rise-fall) as showing that the speaker is “impressed, perhaps awed,” but also “complacent, self-satisfied, even smug” or “shrugging aside any involvement.” These meanings apply if the Jackknife is used with statements, but change if it is used with *wh*-questions, polar questions, or commands (for early discussions of these issues see Liberman 1975: 132ff. and Pierrehumbert and Hirschberg 1990). As a result of these problems, British school models, though still popular in the UK as a means of teaching intonation to L2 learners, are not widely used in research (Nolan 2022).

As briefly noted in §1, cross-linguistic comparisons also argue against claims that there is a natural or direct relationship between tunes and modality or other grammatical and pragmatic functions. This applies particularly to the expression of focus and utterance modality. Focus is not universally expressed as local pitch expansion followed by post-focus compression, as argued, for example, in Xu (2005). Taiwanese uses duration to mark focus rather than changes in pitch (Pan 2008), as does Canadian English to some extent (Kim and Arnhold, 2024). Further, in many languages, pitch expansion is just an optional manifestation of an overall prosodic reorganization under focus (among many, de Jong 1995 on English, Jun 2005a on Korean, Baltazani 2006b on Greek, Chen 2006, 2010 on Mandarin, Venditti, Maekawa, and Beckman 2008 on Japanese). Similarly, questions do not universally end in rises and statements in falls (as would be expected, following Ohala 1983, Gussenhoven 2004, or Xu 2005). Low or falling intonation is used with questions in typologically diverse languages, including Bengali (Hayes and Lahiri 1991), Hungarian and Romanian (Grice et al. 2000), Chickasaw (Gordon 2005), Greek (Arvaniti et al. 2006a), Bininj Gun-wok (Bishop and Fletcher 2005), several Niger-Congo languages (Rialland 2007), and Russian (Post and Andreeva 2023). Conversely, statements end in rises in many languages, including Bengali (Hayes and Lahiri 1991), Chickasaw (Gordon 2005), West Greenlandic (Arnhold 2014), and many varieties of English, where such rises are often referred to as *uptalk* (among many, see Fletcher, Grabe, and Warren 2005, Ritchart and Arvaniti 2014, Arvaniti and Atkins 2016, Armstrong, Piccinini, and Ritchart 2016, and references therein).

This lack of one-to-one correspondence between intonation meaning and form has been repeatedly noted. For instance, Pike (1945: 23ff.) strongly cautioned against the practice of investigating contour meaning on the basis of grammatical structure. Similar observations were made by Bolinger (1964), Lehiste (1970: 95ff. and references therein), Ladd (1980), Pierrehumbert (1980), 't Hart et al. (1990: ch. 4), and more recently by Cangemi and Grice (2016) and Gryllia et al. (in press). Unfortunately, the practice Pike cautioned against continues, for instance when it is assumed that a language has a “question intonation”, a “focus intonation” and the like; these assumptions imply a one-to-one correspondence between function and tune that has been repeatedly shown to be empirically false.

In conclusion, as shown in §3.1, treating tunes as configurations cannot account for observed and well-documented changes in intonation form. Additionally, cross-linguistic research has confirmed the observation made by many scholars over the past century that the relationship between

meaning and tune is (i) arbitrary across languages and (ii) many-to-many within a given linguistic system. These observations support the view that no useful generalizations about either intonational meaning or intonational form can be made on the basis of contour shape and its relationship to broadly defined functions.

4. Compositional approaches

4.1. Early level-based and compositional models

As mentioned in §2, descriptions of intonation by means of level tones were used extensively by the American structuralists (Pike 1945; Trager and Smith 1951; Hockett 1955). Criticism, however, led to the adoption of configurationalist approaches, most notably by Bolinger (1951).

Analyses of intonation by means of level tones began to re-appear in the 1970s, inspired by research on tone languages. Leben (1973, 1976) first conceptualized tones as segments rather than as distinctive features of *tone-bearing units* (TBUs, typically syllables, vowels, or moras with which tones co-occur in speech). Following this idea, Goldsmith (1976) represented tones as *autosegments*, i.e. as autonomous primitives of phonological structure, analysing English tunes as sequences of H (high) and L (low) level tones. Liberman (1975), on the other hand, proposed an analysis with the traditional four levels (extra-high, high, mid, and low) represented by means of two features, $[\pm\text{high}]$ and $[\pm\text{low}]$, though informally, he also used autosegmental representations closer to those of Leben.

A pivotal point in the understanding of intonation was the analysis of the Swedish tonal system in Bruce (1977). Bruce showed that the difference between the lexical pitch accents of Swedish, *accent I* and *accent II*, lies in the timing of the accentual fall with respect to the accented syllable: early for accent I, late for accent II. This timing difference means that only accent I is truncated in utterance-initial position and may appear as low F0. Further, Bruce showed that the second peak of accent II words, the seemingly erratic presence of which had been a longstanding puzzle in Swedish tonology, is not part of accent II at all, but of the utterance's *intonation*, which he analysed as a sequence of a *sentence accent* and a *terminal juncture*. Thus, Bruce's analysis decomposed tunes into constituent elements and showed that lexical and postlexical components (lexical pitch accents and intonation, respectively) concatenate, without one type being superimposed on the other (as in Fujisaki and Hirose 1984). Finally, Bruce demonstrated the importance of *turning points*, F0 minima and maxima which synchronize with particular elements of the segmental string. Bruce showed that these timing relations between text and tune are regular in production and salient in perception, and that turning points are sufficient for modelling contours, which he treated as composites of turning points and F0 interpolation between them.

4.2 The autosegmental-metrical model of intonational phonology

The insights of Bruce and the early autosegmental accounts were applied to the analysis of English intonation in Pierrehumbert (1980). Her model was further developed in Beckman and Pierrehumbert (1986) and Pierrehumbert and Beckman (1988) and is the basis of the approach to intonation referred to as AM (for overviews see Ladd 2008, Arvaniti, 2022, and Arvaniti et al. to appear). The moniker *autosegmental-metrical*, first used by Ladd (1996), refers to the distinction made in AM between the tune, represented as a string of level tones treated as *autosegments*, and the metrical (or *prosodic*) structure, which reflects phrasing and relative constituent strength. Thus, the term AM serves to highlight the fact that both the metrical structure and the string of tonal autosegmental are required for the phonological representation and analysis of intonation (on the importance of this separation, see Ladd and Arvaniti 2023).

In AM, the intonation primitives are H and L tones, a binary that avoids Bolinger's criticism of models with multiple levels. Tunes are represented as strings of Hs and Ls and their combinations. Tones associate with the *segmental* string indirectly, via associations with positions in the metrical structure, namely constituent heads (if available) and boundaries. Tones (and composites of tones) that associate with the (heads of) feet are known as *pitch accents* and are notated with an asterisk (e.g. H*, L+H*). Tones associated with phrasal boundaries are known as *boundary tones* and notated with a percentage sign (e.g. H%). Tones that appear between the boundary tone and the preceding *nuclear accent* (the last pitch accent of an utterance, often referred to in other literature as *sentence stress*) are known as *phrase accents* (e.g. L-). Phonetically, the H and L tones are realized as *tonal targets*, specific points in the F0 contour defined by their *scaling* (relative F0 value) and *alignment*, i.e. their timing relative to the segmental material they are phonologically associated with. Thus, pitch accents phonetically align with stressed syllables, boundary tones with phrase-final TBUs, while phrase accents spread between the nuclear accent and boundary tone (for additional discussion and rarer patterns, see, inter alia, Arvaniti et al. 2000; Grice et al. 2000; Gussenhoven 2000).

A major difference between AM and other models, such as PENTA, INTSINT, the British school and Fujisaki, is that the purpose of AM representations is not to transcribe the course of F0, but to represent its linguistically significant elements. Consequently, some parts of actual pitch contours are not phonologically specified and also remain unspecified at the phonetic level. Empirical evidence for this position was provided by Pierrehumbert and Beckman (1988: ch. 2), who showed that the course of F0 between the initial phrasal H and final L% of Japanese accentual phrases without a lexical pitch accent is a fall that varies in steepness depending on the number of moras between the two tones. This variability in slope is best explained by specifying only the H and L targets and interpolating between them; at the same time, the observed pattern is incompatible with fully specified representations, whether these are present from the start or determined at a later stage through tone spreading.

AM's departure from the tradition of treating pitch contours as the objects of representation successfully addresses one of the main points raised in §1: how to reach significant generalizations over seemingly disparate contours while understanding and predicting their differences in realization. As an illustration, the contours in Figure 32.1 are both phonologically represented as a sequence of a L*+H pitch accent associated with and realized on the stressed syllable of the metrically strongest word in the utterance, and a L- phrase accent followed by a H% boundary tone; these two *edge tones* associate with the intermediate phrase (ip) and intonation phrase (IP) right boundary, respectively (see Figure 32.5). Thus, the fall from the H of L*+H to the following L- is epiphenomenal, the result of interpolation between the accentual H and phrasal L tone. Further, by using the same phonological representation for the two contours in Figure 32.1, AM captures the fact that they are both instantiations of the same melody, thereby generalizing beyond surface form (unlike the British school description of the same tunes as simple and compound fall-rises, discussed in §3.1). Relying on the same representation also accounts for their differences, since by phonologically representing only specific pitch points and decoupling them from the segmental string reflects the observation that, in intonation (as in lexical tonal systems too), the number of tones need not match the number of TBUs. In *Me?! in Figure 32.1*, all three tonal events must co-occur with the only syllable of the utterance, a situation known as *tonal crowding* (see Roettger and Grice 2019 for an overview). Tonal crowding leads to the lengthening of *me* (720 ms) and the swift movement from one tonal target to the next. In *A ballgown designer?! on the other hand*, there are more TBUs than tones: L*+H is associated with *ball*, the metrically strongest syllable in the utterance, and co-occurs with it. The H% is realized as a rise on the last TBU, while the L- phrase accent spreads between the L*+H and H%, accounting for the fall and low-level F0 stretch (for details, see Grice et al.

2000, Barnes et al. 2006). A similar analysis applies to the Greek tune in Figure 32.2 (Baltazani 2006a).

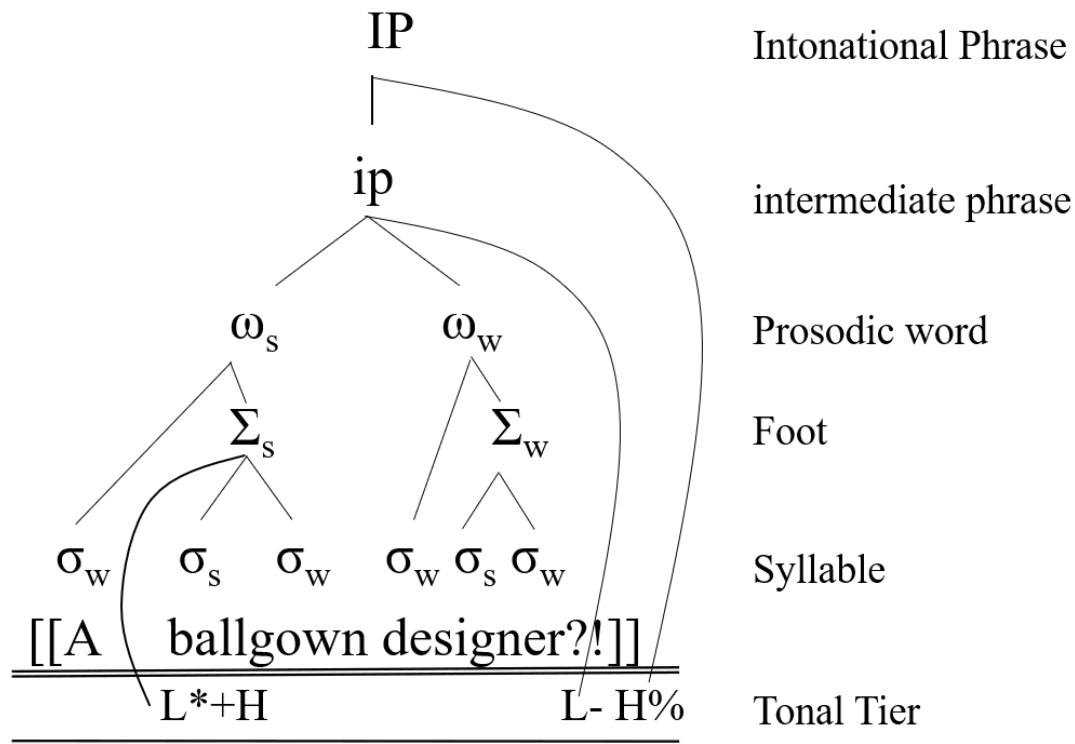


Figure 32.5 Schematic representation of the metrical structure and AM representation of the tune shown in Figure 32.1b.

In the past 45 years, AM has been applied, with various modifications, to scores of languages with typologically distinct prosodic systems (see Jun 2005b, 2014, and Gussenhoven and Chen 2020 for surveys). In addition to “intonation languages” (languages like English that have stress and use F0 only postlexically), AM analyses have been developed for languages without stress (e.g. Jun 2005a on Korean), languages with lexical tone (e.g. Morén and Zsiga 2006 on Thai), and languages with lexical pitch accent (e.g. Elordieta and Hualde 2014 on Basque). This host of studies has largely supported the major tenets of AM.

An important tool in developing these analyses has been the notion of *segmental anchoring*, viz. the idea that tonal targets are stably and invariably aligned with specific positions in the segmental string, such as the accented vowel onset. Segmental anchoring was proposed by Ladd and colleagues (e.g. Ladd et al. 1999) based on the results of Arvaniti et al. (1998), who found such stable alignment for the L and H tones of the Greek L*+H accent (see §2). Segmental anchoring led to invariance being used as a criterion for determining underlying tones, on the assumption that invariable tonal targets reflect tones that should be phonologically represented, while more variable targets do not (e.g. Arvaniti and Ladd 2009).

Strong versions of segmental anchoring have met with criticism, as invariance is statistically impossible to prove and phonetically unrealistic (see Arvaniti et al. 2024 for a discussion). Factors shown to affect alignment include tonal crowding (e.g. Arvaniti and Ladd 2009 on Greek), speaking rate (e.g. Prieto and Torreira 2007 on Spanish), TBU phonological weight and segment identity (Ladd, Mennen and Schepman 2000 on Dutch; Grice, Ridouane, and Roettger, 2015 on Tashlhiyt Berber), and dialectal differences (e.g. Arvaniti and Garding 2007 on American English, Ladd et al. 2009 on British English). Additionally, tones may not be realized as highly localized minima and maxima but as stretches or *plateaux* (Knight and Nolan 2006, and Knight 2008, on British English; D’Imperio et al. 2010 on Italian and German), while accentual pitch peaks reflecting underlying H tones show *peak delay* (among many, Silverman and Pierrehumbert 1990, on H* pitch accents in American English, Arvaniti et al. 1998, on L*+H pitch accents in Greek). As a result of this body of research, it is now recognized that invariance need not imply strict synchronization with the associated TBU; additionally, invariance does not preclude variation driven by linguistic context (what Arvaniti and Ladd, 2009, dubbed *lawful variability*), and cannot possibly preclude the type of random phonetic variability that is part and parcel of speech production.

Further, recent alternatives to measuring the scaling and alignment of tonal targets indicate that targets are not always sufficient for describing intonation *phonetics* and the processing of intonation categories (for a discussion see Arvaniti et al. to appear). Barnes et al. (2012) developed the Tonal Center of Gravity (TCoG) measure to quantify the perceptual differences between concave and convex accents (such as the American English L*+H and L+H*, respectively), the difference between which is hard to capture by relying on just underlying H and L tones and their phonetic targets. Arvaniti et al. (2024) found that the F0 of unstressed syllable varies depending on the following accent: in their data, the F0 on unstressed syllables preceding (low-scaled) H*s was low, while similar syllables preceding (high-scaled) H*+L accents had high F0. Arvaniti et al. (2024) argue that such differences reflect cue redundancy, an observation also supported by studies showing that pitch accent realization involves not only F0 but additional cues, such as changes in duration, amplitude, and voice quality (Arvaniti, Žygis, and Jaskuła 2017 on Polish; Hu and Arvaniti 2024 on Greek). Arvaniti et al. (2024) also show that such additional cues can be in a trading relation with F0: in their study, the lack of a clear F0 dip for L+H* was compensated for by longer accented vowel duration. Taken together, these findings indicate that intonation events behave much like segments, in that their phonetics is characterized by cue redundancy and trading (Arvaniti et al. 2024).

So far, the abovementioned methodological shifts away from tonal target measurement have not led to radical revisions of the basic understanding of AM from the perspective of phonology. Indeed, recent studies appear to support it. For instance, dynamical systems modelling used by Iskarous et al. (2024) showed that the accentual categories posited by AM for American English emerge naturally by applying different settings to a small set of parameters. Nevertheless, some caution is needed when applying newer methodologies, such as Functional Principal Component Analysis (Arvaniti et al. 2024), Generalized Additive (Mixed) Models (Steffman and Cole 2024), and clustering methods (Kaland and Grice 2024). Although these methods were applied to comparable stretches of tunes in the aforementioned studies, using them to model or compare tunes of widely varying lengths can obscure important distinctions between linguistically relevant regions and interpolated stretches. Doing so runs the risk of reverting to configurational approaches which, as discussed in §3.1, are not observationally adequate.

The above warning should not be interpreted to mean that there is no room for debate. Ladd (2008, ch. 8) suggests that abstract tunes unconnected to metrical structure – what we could call configurations – may be part of the mental representation of intonation, allowing for flexibility in

how a tune is metrically interpreted (cf. Torreira and Grice 2018). Recourse to such abstract configurations should not be surprising during processing, especially for frequent tunes used with specific expressions (Calhoun and Schweitzer 2013), as holistic speech processing is known to take place (Hawkins 2003). Additionally, there may be reason to believe that tunes are not as fine-grained as AM models posit; e.g. Cole et al. (2023) used clustering on tunes produced by participants who imitated auditory stimuli emulating tunes derived from Pierrehumbert (1980) and argued that “a binary pitch accent contrast {H*, L*} and a maximally ternary {H%, M%, L%} boundary tone contrast” may be sufficient for describing the tunes used in English. This inventory is much smaller than that of Pierrehumbert (1990) which included seven pitch accents, two phrase accents, and two boundary tones.

Overall, however, the AM approach to intonation avoids several pitfalls of previous analyses. First, by formally separating stress from intonation it provides a clear mechanism for their interaction (for a discussion see Ladd and Arvaniti 2023). Second, by making explicit the separation between the phonetics and phonology of intonation and by recognizing that phonological representations need not account for the entire course of F0 of a given tune, AM provides a parsimonious and elegant way of capturing both significant phonological generalizations and differences in the realization of tunes. Recent research has not significantly altered these essential points, though it has provided both a richer set of methodologies for empirical research and a more nuanced view—on the one hand, of the phonetics of intonation, and on the other, of its relation to phonological representation.

4.3 AM and meaning

Several formal analyses have relied on AM to understand the role of intonation in encoding grammatical functions, most particularly focus and information structure (Steedman 2000, 2014, Büring 2007, among others). More generally, however, AM has not grappled as systematically with meaning as with intonation form (for overviews of intonation meaning from an AM perspective, see Prieto 2015, and Westera, Goodhue, and Gussenhoven 2020). Consequently, many AM analyses of intonation systems rely mostly on phonetics to determine phonological representations.

A major principle of existing AM-based models of intonation meaning is that the meaning of tunes is compositional (Pierrehumbert and Hirschberg 1990; Portes and Beyssade 2015). The advantage of compositional approaches is that they can account for similarities in the meaning conveyed by identical components of different tunes, such as the use of the same pitch accent with different phrasal tones. Further, since no natural relationship between tones and meaning is assumed, compositional models are easier to adapt to different linguistic systems and better suited to account for the variability of natural speech (see, e.g., Arvaniti et al. to appear, on the pragmatics of wh-questions in Greek). Finally, the interpretation of the tune is probabilistic and at any given time will depend, at least in part, on world choice and context (Calhoun 2010, Kurumada and Roettger 2022).

These principles are outlined in Pierrehumbert and Hirschberg (1990), a model of intonation meaning for English, based on the analysis of Beckman and Pierrehumbert (1986). According to Pierrehumbert and Hirschberg (1990), each pitch accent, phrase accent, and boundary tone is a morpheme with its own pragmatic meaning; the meaning conveyed by an entire tune is compositional and depends on contributions from all components in combination with lexical and propositional semantics, world knowledge and pragmatics. As an illustration, H* L- L%, often used with English statements, is not to be interpreted as “S (the speaker) believes x”; rather, the speaker’s belief in x is to be inferred from all the tones that make up a tune and the context in which they are used. This approach can account for the different interpretation of the same tune in *That’s twenty*

dollars and *That's really awesome* in Figure 32.3, briefly discussed in §1. Simplifying somewhat, the downstepped !H* accent (in Pierrehumbert and Hirschberg 1990 represented as H*+L) on *dollar* and *awesome* implies that this item should be inferable by the hearer. This is expected for *dollars* in a context in which dollars are the default currency, but when used with *awesome* the inferred predictability implies that the speaker is either formulaically using an expression associated with excitement (and therefore is not excited) or deliberately uses this accent to convey sarcasm.

Though the principles outlined above are largely supported by empirical studies (e.g. Arvaniti et al. to appear), much remains to be done, as models like Pierrehumbert and Hirschberg (1990) and Portes and Beyssade (2015) have not been extensively tested on naturally occurring data, while experimental studies suggest that some positions may require revision. One such area of disagreement concerns prenuclear accents: contra Pierrehumbert and Hirschberg (1990), Calhoun (2010) has argued that prenuclear accents in English are metrically motivated and do not contribute to meaning (see Buring 2007 for similar arguments). On the other hand, Petrone and D'Imperio (2011) and Baltazani et al. (2015) report that prenuclear accents in Italian and Greek, respectively, do contribute to the interpretation of the tune. Despite such disagreements, however, it is clear, that crude distinctions such as statement vs. question intonation are not sufficient to explain intonational meaning and that a formal, compositional understanding is likely to be more successful.

5. Conclusion

The original debate about levels vs. configurations (framed as such by Bolinger 1951) focused on the issue of whether melodies are gestalts or should be seen as composites of primitives. Independently of this distinction, Bolinger's views, espoused by many before him and since, are based on the idea that intonational contours should be represented in their entirety, either as a series of primitives or as an undulating line. Current understanding suggests that couching the problem in these terms is misleading, as neither type of representation is likely to be correct: as shown, gestalt approaches cannot account in a satisfactory manner for either intonational meaning or intonational form; yet representations that fully specify the course of F0, either in terms of dynamic tones or in terms of levels, do not fare much better. Due to the particularities of intonation, especially the fact that its realization depends on the metrical structure of the utterance with which it co-occurs, significant generalizations about melodies and their phonetic variation are best captured if it is recognized that only certain parts of F0 contours are linguistically relevant and should be phonologically represented. Finally, empirical evidence as well as considerations of representational parsimony strongly suggest that these linguistically relevant aspects of F0 contours are best represented as levels rather than movements.

Acknowledgements

I thank Christina Lee and Tara Boswell for providing the American English examples in Figures 32.1 and 32.3, respectively. The revision of this chapter is part of a project that has received funding from the European Research Council (ERC) under the European Union's Horizon 2020 research and innovation programme (Grant agreement No. ERC-ADG-835263 to Amalia Arvaniti). This support is hereby gratefully acknowledged.

Amalia Arvaniti bio

Amalia Arvaniti is the Professor of English Language and Linguistics at Radboud University, the Netherlands. Since receiving her Ph.D. from the University of Cambridge she has held academic appointments at Cambridge, Oxford, Edinburgh, and Cyprus, and full professorships at the University of California, San Diego and the University of Kent, UK. Her research focuses on the phonetics and

phonology of cross-linguistic prosody, particularly rhythm and intonation, and the phonetics and phonology of Greek. Her research has been funded by the European Research Council, the Leverhulme Trust, the British Academy, the Academy of Korean Studies, and the USA National Science Foundation.

References

- Armstrong, Liliás E., and Ida C. Ward. *Handbook of English intonation*. 1926. Leipzig, Berlin: B. G. Teubner.
- Armstrong, Meghan E., Page Piccinini, and Amanda Ritchart. 2016. Non-question rises in narratives produced by mothers and daughters. *Proc. Speech Prosody 2016*, 123–127, doi: 10.21437/SpeechProsody.2016–26
- Arnhold, Anja. 2014. Prosodic structure and focus realization in West Greenlandic. In S. Jun (ed), *Prosodic Typology II: The Phonology of Intonation and Phrasing*, 216–251. Oxford: Oxford University Press.
- Arvaniti, Amalia. 2012. Prosodic representations. Segment-to-tone association. In Cohn et al. (2012), pp. 265–275.
- Arvaniti, Amalia. 2016. Analytical decisions in intonation research and the role of representations: Lessons from Romani. *Laboratory Phonology* 7(1). 1–43. <https://doi.org/10.5334/labphon.14>
- Arvaniti, Amalia. 2022. The Autosegmental-Metrical Model of Intonational Phonology. In Jonathan Barnes, and Stefanie Shattuck-Hufnagel (eds), *Prosodic Theory and Practice*, 25–63. Cambridge, MA: The MIT Press. <https://doi.org/10.7551/mitpress/10413.003.0004>
- Arvaniti, Amalia, and Madeleine Atkins. 2016. Uptalk in Southern British English. *Proc. Speech Prosody 2016*. 153–157, doi: 10.21437/SpeechProsody.2016-32
- Arvaniti, Amalia, and Mary Baltazani. 2005. Intonational analysis and prosodic annotation of Greek spoken corpora. In Jun (2005b). 84–117.
- Arvaniti, Amalia, Stella Gryllia, and Mary Baltazani. To appear. The complex relationship between the tunes and pragmatics of Greek wh-questions. In Regine Eckardt, George Walkden, and Nicole Dehé (eds), *The Handbook of Noncanonical Questions*. Oxford University Press.
- Arvaniti, Amalia, and Gina Garding. 2007. Dialectal variation in the rising accents of American English. In Jennifer Cole, and José Ignacio Hualde (eds), *Papers in Laboratory Phonology* 9. 547–576. Berlin, and New York: Mouton de Gruyter.
- Arvaniti, Amalia, Martine Grice, Mariapaola D’Imperio. To appear. Advancements of phonetics in the 21st century: Intonation. *Journal of Phonetics*.
- Arvaniti, Amalia, Argyro Katsika, and Na Hu. 2024. Variability, overlap, and cue trading in intonation. *Language* 100(2). 265–307.
- Arvaniti, Amalia, and D. Robert Ladd. 2009. Greek wh-questions and the phonology of intonation. *Phonology* 26. 43–74.
- Arvaniti, Amalia, D. Robert Ladd, and Ineke Mennen. 1998. Stability of tonal alignment: The case of Greek prenuclear accents. *Journal of Phonetics* 26. 3–25.

- Arvaniti, Amalia, D. Robert Ladd, and Ineke Mennen. 2000. What is a starred tone? Evidence from Greek. In Broe, and Pierrehumbert (2000). 119–131.
- Arvaniti, Amalia, D. Robert Ladd, and Ineke Mennen. 2006a. Phonetic effects of focus and “tonal crowding” in intonation: Evidence from Greek polar questions. *Speech Communication* 48. 667–696.
- Arvaniti, Amalia, Marzena Żygis, and Marek Jaskuła. 2017. The phonetics and phonology of the Polish calling melodies. *Phonetica* 73(3–4). 338–361.
- Baltazani, Mary. 2006a. Intonation and pragmatic interpretation of negation in Greek. *Journal of Pragmatics* 38(10). 1658–1676. DOI: 10.1016/j.pragma.2005.03.016
- Baltazani, Mary. 2006b. Focusing, prosodic phrasing, and hiatus resolution in Greek. In Louis Goldstein, Douglas Whalen, and Catherine T. Best (eds), *Laboratory phonology 8*. 473–493. Berlin, and New York: Mouton de Gruyter.
- Baltazani, Mary, Stella Gryllia, and Amalia Arvaniti. 2020. The intonation and pragmatics of Greek wh-questions. *Language and Speech* 63(1): 56–94. DOI: 10.1177/0023830918823236.
- Baltazani, Mary, Evia Kainada, Angelos Lengeris, and Katerina Nicolaidis. 2015. The prenuclear field matters: questions and statements in Standard Modern Greek. *Proceedings of the 18th International Congress of Phonetic Sciences*. <https://www.internationalphoneticassociation.org/icphs-proceedings/ICPhS2015/Papers/ICPHS0941.pdf>
- Barnes, Jonathan, and Stefanie Shattuck-Hufnagel, eds. 2022. *Prosodic Theory and Practice*. Cambridge, MA: The MIT Press.
- Barnes, Jonathan, Stefanie Shattuck-Hufnagel, Alejna Brugos, and Nanette Veilleux. 2006. The domain of realization of the L-phrase tone in American English. *Speech Prosody* 2006. 10.21437/SpeechProsody.2006–75.
- Barnes, Jonathan, Nanette Veilleux, Alejna Brugos, and Stefanie Shattuck-Hufnagel. 2012. Tonal Center of Gravity: A global approach to tonal implementation in a level-based intonational phonology. *Laboratory Phonology* 3(2). 337–383.
- Beckman, Mary E., and Janet B. Pierrehumbert. 1986. Intonational structure in Japanese and English. *Phonology Yearbook* 3. 255–309.
- Bishop, Judith, and Janet Fletcher. 2005. Intonation in six dialects of Biniŋ Gun-wok. In Jun (2005b). 331–361.
- Bolinger, Dwight L. 1951. Intonation: Levels versus configurations. *Word* 7. 199–210.
- Bolinger, Dwight L. 1964. Around the edge of language: Intonation. *Harvard Educational Review* 34. 282–293. Reprinted in Bolinger (1972). 19–29.
- Bolinger, Dwight L. (ed). 1972. *Intonation: Selected readings*. Harmondsworth: Penguin.
- Bolinger, Dwight L. 1986. *Intonation and its parts: Melody in spoken English*. London: Arnold.
- Broe, Michael B., and Janet B. Pierrehumbert (eds). 2000. *Papers in laboratory phonology V: Acquisition and the lexicon*. Cambridge: Cambridge University Press.
- Bruce, Gösta. 1977. *Swedish word accents in sentence perspective*. Lund: Gleerup.

- Büring, Daniel. 2007. Intonation, semantics and information structure. In Gillian Ramchand, and Charles Reiss (eds), *The Oxford handbook of linguistic interfaces*. 445–473. Oxford: Oxford University Press.
- Calhoun, Sasha. 2010. The centrality of metrical structure in signaling information structure: A probabilistic perspective. *Language* 86. 1–42.
- Calhoun, Sasha, and Antje Schweitzer. 2013. Can intonation contours be lexicalised? Implications for discourse meanings. In Gorka Elordieta, and Pilar Prieto (eds), *Prosody and Meaning*. 271–327. De Gruyter Mouton. <https://doi.org/10.1515/9783110261790.271>.
- Calhoun, Sasha, Paul Warren, Joy Mills, and Jemima Agnew. 2024. Socialising the Frequency Code: Effects of gender and age on iconic associations of pitch. *Journal of the Acoustical Society of America* 156(5): 3183–3203. <https://doi.org/10.1121/10.0034354>.
- Cangemi, Francesco, and Martine Grice. 2016. The importance of a distributional approach to categoriality in autosegmental-metrical accounts of intonation. *Laboratory Phonology* 7(1): 9. <https://doi.org/10.5334/labphon.28>
- Chen, Yiya. 2006. Durational adjustment under corrective focus in Standard Chinese. *Journal of Phonetics* 34(2). 176–201. <https://doi.org/10.1016/j.wocn.2005.05.002>
- Chen, Yiya. 2010. Post-focus F0 compression: Now you see it, now you don't. *Journal of Phonetics* 38(4). 517–525. <https://doi.org/10.1016/j.wocn.2010.06.004>
- Cohn, Abigail, Cécile Fougeron, and Marie Huffman (eds). 2012. *The Oxford Handbook of Laboratory Phonology*. Oxford: Oxford University Press.
- Cole, Jennifer, Jeremy Steffman, Stefanie Shattuck-Hufnagel, and Sam Tilsen. 2023. Hierarchical distinctions in the production and perception of nuclear tunes in American English. *Laboratory Phonology* 14(1). doi: <https://doi.org/10.16995/labphon.9437>
- Cruttenden, Alan. 1997. *Intonation*. Cambridge: Cambridge University Press.
- Crystal, David. 1972. The intonation system of English. In Bolinger (1972). 110–136.
- de Jong, Kenneth J. 1995. The supraglottal articulation of prominence in English: Linguistic stress as localized hyperarticulation. *Journal of the Acoustical Society of America* 97. 491–504.
- Dilley, Laura C., and Meredith Brown. 2007. Effects of pitch range variation on f0 extrema in an imitation task. *Journal of Phonetics* 35(4). 523–551.
- Dilley, Laura C., D. Robert Ladd, and Astrid Schepman. 2005. Alignment of L and H in bitonal pitch accents: testing two hypotheses. *Journal of Phonetics* 33(1). 115–119.
- D'Imperio, Mariapaola, and David House. 1997. Perception of questions and statements in Neapolitan Italian. *Proceedings of the 5th European Conference on Speech Communication and Technology (Eurospeech 1997)*. Vol. 1, 251–254. Rhodes, Greece.
- D'Imperio, Mariapaola, Barbara Gili Fivela, and Oliver Niebuhr. 2010. Alignment perception of high intonational plateaux in Italian and German. *Proc. Speech Prosody 2010*, paper 186, doi: 10.21437/SpeechProsody.2010–153
- Elordieta, Gorka, and José Ignacio Hualde. 2014. Intonation in Basque. In Jun (2014). 405–463.

- Fletcher, Janet, Esther Grabe, and Paul Warren. 2005. Intonational variation in four dialects of English: The high rising tune. In Jun (2005b). 390–409.
- Fougeron, Cécile, and Sun-Ah Jun. 1998. Rate effects on French intonation: Prosodic organization and phonetic realization. *Journal of Phonetics* 26. 45–69.
- Frota, Sónia. 2002. Tonal association and target alignment in European Portuguese nuclear falls. In Carlos Gussenhoven, and Natasha Warner (eds), *Laboratory phonology* 7. 387–418. Berlin, and New York: Mouton de Gruyter.
- Fujisaki Hiroya, and Keikichi Hirose. 1984. Analysis of voice fundamental frequency contours for declarative sentences of Japanese. *Journal of the Acoustical Society of America* 5(4). 233–242.
- Gårding, Eva. 1983. A generative model of intonation. In Anne Cutler, and D. Robert Ladd (eds), *Prosody: Models and measurements*. 11–25. Berlin: Springer.
- Goldsmith, John A. 1976. Autosegmental phonology. Ph.D. dissertation, MIT. Published 1979, New York: Garland.
- Goodhue, Daniel, Lyana Harrison, Yuen Tung Clementine Siu, and Michael Wagner. 2015. Toward a bestiary of English intonational tunes. In C. Hammerly, and B. Prickett (eds), *Proceedings of the 46th Conference of the North Eastern Linguistic Society (NELS)*, Concordia University. 311–320.
- Gordon, Matthew. 2005. Intonational phonology of Chickasaw. In Jun (2005b). 301–330.
- Grabe, Esther, Greg Kochanski, and John Coleman. 2003. Quantitative modelling of intonational variation. *Proceedings of Speech Analysis and Recognition in Technology, Linguistics and Medicine 2003*. Available at <http://www.phon.ox.ac.uk/oxigen/publications.php> (accessed August 3, 2010).
- Grice, Martine, D. Robert Ladd, and Amalia Arvaniti. 2000 On the place of phrase accents in intonational phonology. *Phonology* 17(2). 143–185.
- Grønnum, Nina. 2022. Modeling Danish intonation. In J. Barnes, and S. Shattuck-Hufnagel (eds). 85–116.
- Gussenhoven, Carlos. 2000. The boundary tones are coming: On the non-peripheral realization of boundary tones. In Broe, and Pierrehumbert (2000). 132–151.
- Gussenhoven, Carlos. 2004. *The phonology of tone and intonation*. Cambridge: Cambridge University Press.
- Gussenhoven, Carlos, and Aojun Chen (eds). 2020. *The Oxford Handbook of Language Prosody*. Oxford University Press.
- Halliday, M. A. K. 1970. *A course in spoken English: Intonation*. Oxford: Oxford University Press.
- Hart, Johan 't, René Collier, and Antonie Cohen. 1990. *A perceptual study of intonation: An experimental-phonetic approach to speech melody*. Cambridge: Cambridge University Press.
- Hawkins, Sarah. 2003. Roles and representations of systematic fine phonetic detail in speech understanding. *Journal of Phonetics* 31(3–4). 373–405.
- Hayes, Bruce, and Aditi Lahiri. 1991. Bengali intonational phonology. *Natural Language and Linguistic Theory* 9. 47–96.

- Hirschberg, Julia. 2004. Pragmatics and intonation. In Larry R. Horn, and Gregory Ward (eds), *The Handbook of Pragmatics*. 515–537. Wiley. <https://doi.org/10.1002/9780470756959.ch23>
- Hirschberg, Julia, and Gregory Ward. 1992. The influence of pitch range, duration, amplitude and spectral features on the interpretation of the rise–fall–rise intonation contour in English. *Journal of Phonetics* 20. 241–251.
- Hirst, Daniel, and Albert Di Cristo. 1998. A survey of intonation systems. In Daniel Hirst, and Albert Di Cristo (eds), *Intonation systems: A survey of twenty languages*. 1–44. Cambridge: Cambridge University Press.
- Hockett, Charles F. 1955. *A manual of phonology*. Baltimore: Waverly Press.
- Hu, Na, and Amalia Arvaniti. 2024. Individual variability in the use of tonal and non-tonal cues in intonation. *Journal of the Acoustical Society of America Express Letter* 4(9). 095203. <https://doi.org/10.1121/10.0028613>
- Iskarous, Khalil, Jennifer Cole, and Jeremy Steffman. 2024. A minimal dynamical model of Intonation: Tone contrast, alignment, and scaling of American English pitch accents as emergent properties. *Journal of Phonetics* 104: 101309.
- Jones, Daniel. 1909. *Intonation curves. A collection of phonetic texts, in which intonation is marked throughout by means of curved lines on a musical stave*. Leipzig, Berlin: B. G. Teubner.
- Jones, Daniel. 1932. *An outline of English phonetics*. 3rd edn. Cambridge: Cambridge University Press.
- Jones, Daniel. 1972. *An outline of English phonetics*. 9th edn. Cambridge: Heffer and Sons.
- Jun, Sun-Ah. 2005a. Korean intonational phonology and prosodic transcription. In Jun (2005b). 201–229.
- Jun, Sun-Ah (ed). 2005b. *Prosodic typology: The phonology of intonation and phrasing*. Oxford: Oxford University Press.
- Jun, Sun-Ah (ed). 2014. *Prosodic Typology II: The Phonology of Intonation and Phrasing*. Oxford University Press.
- Jun, Sun-Ah, and Janet Fletcher. 2014. Methodology of studying intonation: from data collection to data analysis. In S. Jun (ed). 493–519.
- Kaland, Constantine, and Martine Grice. 2024. Exploring and explaining variation in phrase-final f0 movements in spontaneous Papuan Malay. *Phonetica* 81 (3). <https://doi.org/10.1515/phon-2023-0031>
- Kim, Jiseung, and Anja Arnhold. 2024. Phonetic and phonological aspects of prosodic focus marking in Canadian English. *Laboratory Phonology* 15(1). doi: [10.16995/labphon.9316](https://doi.org/10.16995/labphon.9316)
- Jiseung Kim, Na Hu, Stella Gryllia, Riccardo Orrico, and Amalia Arvaniti. 2024. Delineating H* and L+H* in Southern British English. *Speech Prosody* 2024. doi: 10.21437/SpeechProsody.2024-239
- Knight Rachael-Anne. 2008. The shape of nuclear falls and their effect on the perception of pitch and prominence: peaks vs. plateaux. *Language and Speech* 51(3). 223–244.

- Knight, Rachael-Anne, and Francis Nolan. 2006. The effect of pitch span on intonational plateau. *Journal of the International Phonetic Association* 36. 21–38.
- Kurumada, Chigusa, and Timo B. Roettger. 2022. Thinking probabilistically in the study of intonational speech prosody. *WIREs Cognitive Science* 13(1), Article e1579. <https://doi.org/10.1002/wcs.1579>
- Ladd, D. Robert. 1980. *The structure of intonational meaning: Evidence from English*. Bloomington: Indiana University Press.
- Ladd, D. Robert. 1996. *Intonational phonology*. Cambridge: Cambridge University Press.
- Ladd, D. Robert. 2008. *Intonational phonology*. 2nd edn. Cambridge: Cambridge University Press.
- Ladd, D. Robert, and Amalia Arvaniti. 2023. Prosodic prominence across languages. *Annual Review of Linguistics* 9. 171–193.
- Ladd, D. Robert, Dan Faulkner, Hanneke Faulkner, and Astrid Schepman. 1999. Constant “segmental anchoring” of F0 movements under changes in speech rate. *Journal of the Acoustical Society of America* 106(3). 1543–1554.
- Ladd, D. Robert, Ineke Mennen, and Astrid Schepman. 2000. Phonological conditioning of peak alignment in rising pitch accents in Dutch. *Journal of the Acoustical Society of America* 107. 2685–2696.
- Ladd, D. Robert, Astrid Schepman, Laurence White, Louise May Quarmby, and Rebekah Stackhouse. 2009. Structural and dialectal effects on pitch peak alignment in two varieties of British English. *Journal of Phonetics* 37. 145–161.
- Leben, William R. 1973. *Suprasegmental phonology*. Ph.D. dissertation, MIT.
- Leben, William R. 1976. *The tones of English intonation*. *Linguistic Analysis* 2. 69–107.
- Lehiste, Ilse. 1970. *Suprasegmentals*. Cambridge, MA: MIT Press.
- Lieberman, Mark. 1975. *The intonational system of English*. Ph.D. dissertation, MIT.
- Morén, Bruce, and Elizabeth C. Zsiga. 2006. The lexical and postlexical phonology of Thai tones. *Natural Language and Linguistic Theory* 24. 113–178.
- Nolan, Francis. 2022. The rise and fall of the British School of intonation analysis. In J. Barnes, and S. Shattuck-Hufnagel (eds). 31–349.
- O’Connor, J. D., and G. F. Arnold. 1973. *Intonation of colloquial English*. London: Longman.
- Ohala, John J. 1983. Cross-language view of pitch: An ethological view. *Phonetica* 40. 1–18.
- Pan, Ho-Hsien. 2008. Focus and Taiwanese unchecked tones. In Chungmin Lee, Matthew Gordon, and Daniel Büring (eds.) *Topic and focus: Cross-linguistic perspectives on meaning and intonation*. 195–213. Dordrecht: Springer.
- Petrone, Caterina, and Mariapaola D’Imperio. 2015. From tones to tunes: Effects of the f0 prenuclear region in the perception of Neapolitan statements and questions. In S. Frota, G. Elordieta, P. Prieto (eds), *Prosodic Categories: Production, Perception and Comprehension*. 207–230. Springer.
- Pierrehumbert, Janet B. 1980. *The phonology and phonetics of English intonation*. Ph.D. dissertation, MIT.

Pierrehumbert, Janet B., and Mary E. Beckman. 1988. *Japanese tone structure*. Cambridge, MA: MIT Press.

Pierrehumbert, Janet B., and Julia Hirschberg. 1990. The meaning of intonational contours in the interpretation of discourse. In Philip R. Cohen, Jerry Morgan, and Martha E. Pollack (eds), *Intentions in communication*. 271–311. Cambridge MA: MIT Press.

Pike, Kenneth L. 1945. *The intonation of American English*. Ann Arbor: University of Michigan Press.

Portes, Cristel, and Claire Beyssade. 2015. Is intonational meaning compositional? *Verbum* 37(2). 207–233.

Post, Margje, and Bistra Andreeva. 2023. Polar question intonation in Russian speech from Moscow and Perm. In Oliver Niebuhr (ed), *Proceedings of the 13th International Conference of Nordic Prosody*. 147–154. Warsaw: Sciendo. <https://doi.org/10.2478/9788366675728-012>

Prieto, Pilar. 2015. Intonational meaning. *Wiley Interdisciplinary Reviews: Cognitive Science* 6. 371–381.

Prieto, Pilar, and Francisco Torreira. 2007. The segmental anchoring hypothesis revisited: Syllable structure and speech rate effects on peak timing in Spanish. *Journal of Phonetics* 35. 473–500.

Redi, Laura. 2003. Categorical effects in the production of pitch contours in English. In M. J. Solé, D. Recasens, and J. Romero (eds), *Proceedings of the 15th International Congress of Phonetic Sciences*. 2921–2924. Barcelona: Causal Productions.

Rialland, Annie. 2007. Question prosody: An African perspective. In Carlos Gussenhoven, and Tomas Riad (eds), *Tones and tunes, Vol. 1: Typological studies in word and sentence prosody*. 35–62. Berlin, and New York: Mouton de Gruyter.

Rietveld, Toni, and Carlos Gussenhoven. 1995. Aligning pitch targets in speech synthesis: Effects of syllable structure. *Journal of Phonetics* 23. 375–385.

Ritchart, Amanda, and Amalia Arvaniti. 2014. The form and use of uptalk in Southern California English. *Speech Prosody* 7. https://www.isca-archive.org/speechprosody_2014/ritchart14_speechprosody.pdf

Roettger, Timo B., and Martine Grice. 2019. The tune drives the text: Competing information channels of speech shape phonological systems. *Language Dynamics and Change* 9(2). 265–98.

Silverman, Kim E. A., and Janet B. Pierrehumbert. 1990. The timing of prenuclear high accents in English. In John Kingston, and Mary E. Beckman (eds), *Papers in laboratory phonology I: Between the grammar and physics of speech*. 72–106. Cambridge: Cambridge University Press.

Steedman, Mark. 2000. Information structure and the syntax–phonology interface. *Linguistic Inquiry* 31. 649–689.

Steedman, Mark. 2014. The surface-compositional semantics of English intonation. *Language* 90(1). 2–57.

Steffman, Jeremy, and Jennifer Cole. 2024. Metrical enhancement in American English nuclear tunes. *Glossa: a journal of general linguistics* 9(1). doi: 10.16995/glossa.15297

Torreira, Francisco, and Martine Grice. 2018. Melodic constructions in Spanish: Metrical structure determines the association properties of intonational tones. *Journal of the International Phonetic Association* 48(1). 9–32.

Torres, Humberto M., Jorge A. Gurlekian, Hansjörg Mixdorff, and Hartmut Pfitzinger. 2014. Linguistically motivated parameter estimation methods for a superpositional intonation model. *EURASIP Journal on Audio, Speech, and Music Processing* 28. <https://doi.org/10.1186/s13636-014-0028-3>

Trager, George L., and Henry L. Smith. 1951. *An outline of English structure*. Norman: Battenburg Press.

Venditti, Jennifer, Kikuo Maekawa, and Mary E. Beckman. 2008. Prominence marking in the Japanese intonation system. In Shigeru Miyagawa, and Mamoru Saito (eds), *The Oxford handbook of Japanese linguistics*. 456–512. Oxford: Oxford University Press.

Ward, Gregory, and Julia Hirschberg. 1985. Implicating uncertainty: The pragmatics of fall–rise intonation. *Language* 61. 747–776.

Westera, Matthijs, Daniel Goodhue, and Carlos Gussenhoven. 2020. Meanings of Tones and Tunes. In Gussenhoven, and Chen (eds). 442–453. Oxford University Press.

Xu, Yi. 2005. Speech melody as articulatorily implemented communicative functions. *Speech Communication* 46. 220–251.

Xu, Yi, Santitham Prom-on, and Fang Liu. 2022. The PENTA model: Concept, use, and implications. In Barnes, and Shattuck-Hufnagel (eds). 377–407. The MIT Press.